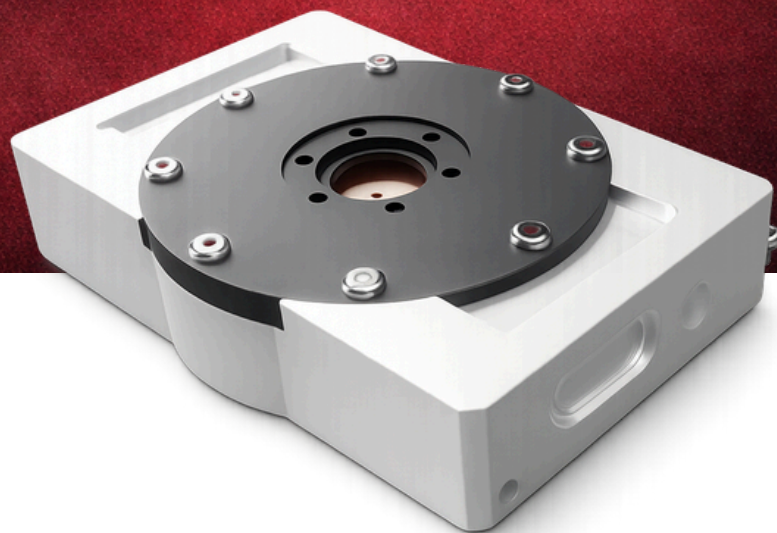


Cryogenic NanoStage

Shielded Signal Integrity for Quantum & Cryogenic Measurements



Ultra-Precise Rotation.
Cryogenic Ready.
Built for Performance.

The Nano Stage is a precision thermal control platform engineered for advanced microscopy, spectroscopy, and materials characterization applications. The system enables stable operation from cryogenic temperatures below -195°C up to elevated temperatures, while maintaining exceptional thermal stability and optical accessibility.

Designed for demanding scientific environments, the stage supports vacuum and controlled atmosphere operation and is compatible with a wide range of optical and analytical techniques including confocal microscopy, Raman spectroscopy, and X-ray analysis.

Notes

- Designed for vitrified sample preservation and cryogenic workflows
- Requires proper LN_2 handling and cryogenic safety procedures
- Performance dependent on microscope compatibility and setup

KEY FEATURES

- **Ultra-Wide Operating Temperature Range:**
Supports operation across an extensive temperature range for diverse experimental requirements.
- **High Temperature Stability**
Maintains temperature stability below 0.01°C for precision measurements.
- **Cryogenic Cooling System**
Direct coolant injection enables efficient cryogenic temperature control.
- **Vacuum & Controlled Atmosphere Capability**
Supports vacuum environments and controlled gas atmospheres for advanced experimentation.
- **Gold-Plated OFHC Copper Block**
High thermal conductivity block ensures efficient and uniform heat transfer.
- **Ultra-Thin Optical Window**
Optimized for high numerical aperture imaging with minimal optical distortion.

APPLICATIONS

- Cryogenic materials analysis
- Semiconductor characterization
- Superconducting material studies
- Raman spectroscopy at low temperature
- In-situ thermal microscopy
- Vacuum microscopy experiments
- Quantum material imaging
- Failure analysis
- Biological cryogenic imaging

Cryogenic NanoStage

SPECIFICATIONS

CRYOGENIC PERFORMANCE

Cooling Method	Liquid nitrogen (LN ₂)
Operating Temperature	< -195 °C
Thermal Stability	Maintains vitrified state
Environment	Frost-controlled sealed chamber

POSITIONING SYSTEM

Stage Type	Motorised XY nano-positioning
Encoding	Integrated position encoding
Motion Control	High precision, repeatable
Mapping Capability	Full grid automation supported

SAMPLE HANDLING

Cassette Type	Magnetic self-aligning
Capacity	Up to 3 EM grids per cassette
Supported	Formats EM grids, autogrids, membranes, planchettes, custom
Transfer Method	Cryogenic transfer container

OPTICAL & IMAGING INTEGRATION

Illumination	Integrated LED condenser
Imaging Compatibility	Cryo-fluorescence, Cryo-CLEM, super-resolution
Objective Support	High-NA objectives (microscope dependent)
Drift Control	Optional objective heater

CONTROL & INTERFACE

Manual Control	Joystick touch controller
Computer Interface	USB
Software Integration	Third-party microscopy platforms (LabVIEW, Python, MATLAB)
Automation	Mapping, positioning, repeat imaging

MECHANICAL & SYSTEM INTEGRATION

Mounting	Quick-release microscope integration
Compatibility	Upright research microscopes
Workflow Integration	Plunge freezing, HPF, Cryo-EM pipelines
Maintenance	Integrated bake-out heating cycle

SPECIFICATIONS

Sample Area	30 mm with a gold-coated oxygen-free copper block for high thermal conductivity
Temperature Range	-195°C to 250°C
Temperature Stability	<0.1°C for ultra-precise temperature control
Electrical Connections	Equipped with Lemo connectors and posts
Vacuum Compatibility	Less than 1×10 ⁻⁵ mBar vacuum level
Objective Lens Working Distance	5 mm
Temperature Controller	USB interface, LabVIEW compatible
Variable Heating Rates	Adjustable for different experimental requirements
Quick-Release Gas Valves	For atmospheric control
Glass Window Area	20 mm

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